Light

GCSE Computer Science - Cyber Security Prevention (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the purpose of an automatic account lockout policy?

- ○ **A.** To prevent brute-force attacks by temporarily disabling accounts after multiple failed login attempts
- ○ **B.** To log users out when they forget their passwords
- ○ **C.** To delete accounts that are not used frequently
- ○ **D.** To speed up the login process

Q2 (1 mark)

What is an audit trail in computer security?

- ○ **A.** A physical path through a data centre
- ○ **B.** A chronological record of system activities that tracks who did what and when
- ○ **C.** A method of testing hiking trails with computers
- ○ **D.** A way to speed up computer processing

Q3 (1 mark)

What happens when anti-malware software quarantines a file?

- ○ **A.** The file is isolated in a secure location where it cannot execute or cause harm
- ○ **B.** The file is immediately deleted permanently
- ○ **C.** The file is sent to the software manufacturer
- ○ **D.** The file is shared with all users on the network

Q4 (1 mark)

What does HTTPS provide that HTTP does not?

- ○ **A.** Faster loading of web pages
- ○ **B.** More colourful website designs
- ○ **C.** Encrypted communication between browser and web server
- ○ **D.** Access to more websites

Q5 (1 mark)

Why is it important to install software updates and patches promptly?

- ○ **A.** To change the software's appearance
- ○ **B.** To make the software more expensive
- ○ **C.** Updates always add new games
- ○ **D.** Updates fix known security vulnerabilities that attackers could exploit

Q6 (1 mark)

What is the primary purpose of regular data backups?

- ○ **A.** To enable recovery of data if original files are lost, corrupted, or encrypted by ransomware
- ○ **B.** To make computers run faster
- ○ **C.** To delete old files automatically
- ○ **D.** To share files with competitors

Q7 (1 mark)

Why must old hard drives be securely destroyed rather than simply deleted?

- ○ **A.** Deleted files can often be recovered using specialist software
- ○ **B.** Deleting files makes hard drives heavier

- ○ **C.** Secure destruction is cheaper than deletion
- ○ **D.** Deleted files automatically appear on the internet

Q8 (1 mark)

What is a firewall rule?

- ○ **A.** A physical barrier around the firewall device
- ○ **B.** A specific instruction telling the firewall whether to allow or block certain types of network traffic
- ○ **C.** A legal regulation about firewalls
- ○ **D.** A game played by network administrators

Q9 (1 mark)

What is a password manager?

- ○ **A.** A person employed to remember passwords
- ○ **B.** A device that types passwords automatically
- ○ **C.** Software that securely stores and manages multiple passwords, often generating strong unique passwords
- ○ **D.** A method for sharing passwords with others

Q10 (1 mark)

In symmetric encryption, what is used to both encrypt and decrypt data?

- ○ **A.** The same single key for both encryption and decryption
- ○ **B.** Two different keys - one public and one private
- ○ **C.** No key is required
- ○ **D.** A different key for each character

Q11 (1 mark)

What is the difference between authentication and authorisation?

- ○ **A.** They are exactly the same thing
- ○ **B.** Authentication verifies identity (who you are); authorisation determines access permissions (what you can do)
- ○ **C.** Authentication is for computers; authorisation is for humans
- ○ **D.** Authorisation happens before authentication

Q12 (1 mark)

What is network monitoring used for in cyber security?

- ○ **A.** To make the network cables look tidy
- ○ **B.** To block all network traffic permanently
- ○ **C.** To detect suspicious activity, identify attacks, and track network performance in real time
- ○ **D.** To increase the cost of internet access

Q13 (2 marks)

A company follows the '3-2-1 backup rule': 3 copies of data, on 2 different media types, with 1 copy stored offsite.
Why is this strategy effective?

- ○ **A.** Multiple copies protect against single failures; different media protects against media-specific failures; offsite storage protects against site disasters like fire or flood
- ○ **B.** It uses lucky numbers that prevent data loss
- ○ **C.** It is required by law for all businesses
- ○ **D.** Three copies make files three times smaller

Q14 (2 marks)

Asymmetric encryption uses a pair of keys: a public key and a private key.
How does this solve the key exchange problem of symmetric encryption?

- ○ **A.** Asymmetric encryption does not use any keys
- ○ **B.** Both keys must be kept secret and exchanged privately
- ○ **C.** The keys are the same, just with different names
- ○ **D.** The public key can be freely shared for encryption, but only the private key (kept secret) can decrypt - no secret key exchange is needed

Q15 (2 marks)

Some security software runs suspicious files in a 'sandbox' before allowing them to execute normally.
What is the purpose of sandboxing?

- ○ **A.** To test if files work on beach computers
- ○ **B.** To compress files to make them smaller
- ○ **C.** To isolate and observe suspicious file behaviour in a controlled environment without risking the actual system
- ○ **D.** To make files run faster by removing restrictions

Q16 (2 marks)

A penetration test includes sending fake phishing emails to employees to see who clicks links or provides credentials.
What security aspect does this test?

- ○ **A.** The speed of the email server
- ○ **B.** The colour scheme of company emails
- ○ **C.** The number of employees in the company
- ○ **D.** The human element - employee awareness and susceptibility to social engineering attacks

Q17 (2 marks)

A stateful firewall tracks the state of active network connections.
What advantage does this provide over a simple packet filter?

- ○ **A.** By tracking connection state, it can identify packets that are part of legitimate established connections versus unexpected packets that might be attacks
- ○ **B.** Stateful firewalls are always cheaper
- ○ **C.** Stateful firewalls do not need any rules configured

- ○ **D.** They block all traffic regardless of rules

Q18 (2 marks)

When connecting to a website using HTTPS, the browser checks the site's SSL/TLS certificate.
What does this certificate verify?

- ○ **A.** That the website has good content
- ○ **B.** That the website is who it claims to be, verified by a trusted Certificate Authority
- ○ **C.** That the website will load quickly
- ○ **D.** That the website owner has paid their internet bill

Q19 (2 marks)

An attacker gains access with a low-privilege user account and then exploits a vulnerability to gain administrator access.
What is this technique called?

- ○ **A.** Password guessing
- ○ **B.** Account deletion
- ○ **C.** Network mapping
- ○ **D.** Privilege escalation - moving from lower to higher access levels within a system

Q20 (2 marks)

A data centre has fire suppression systems, temperature monitoring, and backup power generators.
What type of security threat do these measures address?

- ○ **A.** Social engineering attacks
- ○ **B.** Password cracking attempts
- ○ **C.** Environmental threats - fire, overheating, and power failures that could damage equipment or cause data loss
- ○ **D.** Phishing emails

Q21 (2 marks)

Anti-malware software offers both real-time protection and scheduled scans.
Why are both options important?

- ○ **A.** They are identical - only one is needed
- ○ **B.** Real-time protection catches threats as they arrive; scheduled scans detect malware that may have evaded real-time detection or existed before protection was installed
- ○ **C.** Real-time is for day use; scheduled is for night use
- ○ **D.** Scheduled scans make real-time protection unnecessary

Q22 (2 marks)

Security experts often recommend using passphrases like 'correct horse battery staple' instead of complex passwords like 'Tr0ub4dor&3'.
Why might a passphrase be more secure?

- ○ **A.** Passphrases are always shorter than passwords

- B. Computers cannot process words, only symbols
- C. Passphrases do not need to be kept secret
- D. Passphrases are longer, creating more possible combinations, while being easier to remember than random characters

Q23 (2 marks)

After completing a penetration test, the security team produces a detailed report for the client.
What should this report include?

- A. Only positive findings - vulnerabilities should not be documented
- B. Personal information about employees who failed phishing tests
- C. Discovered vulnerabilities, their severity ratings, potential impact, and recommended remediation steps
- D. Passwords discovered during the test for the client to reuse

Q24 (3 marks)

An attacker attempts to breach a company network. The firewall blocks their initial connection attempts. They then try phishing, but anti-malware blocks the payload. Finally, they attempt physical access but are stopped by locked doors and CCTV.
Analyse why this layered approach was effective.

- A. The attacker was simply unlucky - any single measure would have worked alone
- B. Each security layer addressed a different attack vector: firewall blocked network attacks, anti-malware blocked malware delivery,

physical security blocked physical access - the attacker had no unprotected path to succeed

○ **C.** The attacker should have tried harder

○ **D.** Layered security always guarantees 100% protection

Q25 (3 marks)

A company is hit by ransomware. Their backup strategy includes: daily automated backups to network storage, weekly backups to external drives stored offsite, and monthly backups to cloud storage with versioning.
Analyse how this strategy affects their recovery options.

○ **A.** Backups are useless against ransomware

○ **B.** Only cloud backups matter; other backups have no value

○ **C.** The company should pay the ransom instead of using backups

○ **D.** Network backups may be encrypted by ransomware spreading on the network; external drives offsite are protected from network spread; cloud versioning allows recovery of pre-infection versions - multiple options provide resilience

Q26 (3 marks)

A web application stores user passwords. The developer considers: storing passwords in plain text, encrypting passwords with a key stored on the server, or hashing passwords with a unique salt for each user. Analyse which approach provides the best security and why.

○ **A.** Plain text is most secure because nothing can go wrong with decryption

○ **B.** Encryption is best because the key can always decrypt passwords when needed

○ **C.** Salted hashing is most secure: hashes cannot be reversed, salts prevent rainbow table attacks, and even database breaches do not directly expose passwords - the application verifies passwords by comparing hashes

○ **D.** All three approaches are equally secure

Q27 (3 marks)

A security investigation reveals that confidential files were accessed inappropriately. Audit logs show: which user account accessed the files, the exact times of access, what actions were performed, and the IP address of the accessing computer.
Analyse how audit trails support security investigations.

○ **A.** Audit logs are only useful for improving computer performance

○ **B.** Audit trails provide evidence of who, when, what, and where: user identity shows accountability, timestamps establish timelines, action records show what was done, and IP addresses help locate the source - together enabling investigation and accountability

○ **C.** Audit logs should be deleted to save storage space

○ **D.** Only the username matters; other details are irrelevant

Q28 (3 marks)

A company places their public web server in a DMZ (Demilitarised Zone) - a network segment between the external firewall and internal firewall. Analyse how this architecture improves security.

○ **A.** DMZ stands for 'Data Management Zone' and organises files better

○ **B.** DMZ servers run faster because they have their own network

○ **C.** The internal and external networks are combined for efficiency

○ **D.** The DMZ isolates public-facing servers: external firewall allows web traffic to DMZ only; internal firewall blocks DMZ from accessing internal network - even if the web server is compromised, attackers cannot directly reach internal systems

Q29 (3 marks)

A company detects a security breach in progress. Their incident response plan specifies: contain the breach, eradicate the threat, recover systems, and then conduct post-incident review.
Analyse why this ordered approach is important.

○ **A.** The order does not matter - any sequence works equally well

○ **B.** Containment stops ongoing damage, eradication removes the threat completely, recovery restores normal operations, and review identifies lessons to prevent recurrence - each step builds on the previous to resolve the incident systematically

○ **C.** Only recovery matters - other steps waste time

○ **D.** Post-incident review should happen first to understand what to do

Q30 (3 marks)

A company's security team proposes: 20-character minimum passwords changed weekly, biometric scans for every room, and approval workflows for all file access. Staff complain these measures prevent them from working effectively.
Analyse the balance between security and usability.

○ **A.** Security is always more important than usability - staff complaints should be ignored

○ **B.** Usability is always more important than security - all restrictions should be removed

○ **C.** Security and usability are completely unrelated concerns

○ **D.** Excessive security causes workarounds: users write down complex passwords, prop doors open, or share credentials - creating new vulnerabilities. Effective security must balance protection with usability to achieve compliance without creating bypasses

Total: **55** marks • Time: **30 min**

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